Alexandra Ion

Assistant Professor, Interim Associate Director of HCI Undergraduate Programs
Human-Computer Interaction Institute | School of Computer Science | Carnegie Mellon University
407 S Craig St, Pittsburgh, PA 15213, USA

alexandraion@cmu.edu | https://interactive-structures.org/ | google scholar

Alexandra Ion | **Curriculum Vitae**

Research interests: HCI, physical AI, adaptive physical user interfaces, robotic materials, inverse design, digital fabrication.

Human-Centered Physical AI. We research **adaptive systems for physical interaction** that **respond to user needs**. Our research advances the creation of responsive materials and intelligent structures that sense, react, adapt, and communicate with users and the environment. We design computational methods and innovative materials that seamlessly blend computation, fabrication, and robotics. Applications include assistive devices, rehabilitation, architecture, smart homes, and daily activities. We bridge the gap between understanding user needs and adapting to them by developing full-stack research prototypes, incl. software systems and physical interfaces.

Academic Employment

Jan 2021 – present	Assistant Professor, Carnegie Mellon University School of Computer Science, Human-Computer Interaction Institute Robotics Institute (courtesy), Mechanical Engineering (courtesy)
May 2026 – Dec 2026	Interim Associate Director of HCI Undergraduate Programs
Mar 2019 – Nov 2020	Postdoctoral researcher at ETH Zürich Advisor: Prof. Olga Sorkine-Hornung, Interactive Geometry Lab

Educational Background

Sep 2018 – Jan 2019	Research stay at ETH Zürich, Advisor: Prof. Olga Sorkine-Hornung, Interactive Geometry Lab
Jul 2013 – Apr 2019 ***Dr.*** *rer. nat.*	PhD student at Hasso Plattner Institute, University of Potsdam Advisor: Prof. Patrick Baudisch, Human-Computer Interaction Thesis: Metamaterial Devices Committee: Prof. Scott Hudson, Prof. Sriram Subramanian, Prof. Marc Alexa
Oct 2010 - Nov 2012 ***MSc***	University of Applied Sciences Upper Austria, Hagenberg Thesis advisor: Prof. Michael Haller Thesis: Uncovering moving off-view objects on large interactive displays.
May 2012 - Oct 2012	Research stay at University of Waterloo, ON, Canada. Advisors: Prof. Mark Hancock, Prof. Stacey Scott
Oct 2006 - Jul 2009 ***BSc***	University of Applied Sciences Upper Austria, Hagenberg

Publications

PEER-REVIEWED PAPERS

ACM CHI and UIST are premiere venues for HCI, 20-26% acceptance rate.

[33] Objects with Arms: How Task-Relevant Embodiment Shapes Perception in Physical Collaboration
Violet Yinuo Han, Ziru Wei, Aiden Yiliu Li, Chris Wu, Alexandra Ion
In Proceedings of the 2026 IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2026), Aug 24 - Aug 28, 2026, Kitakyushu, Japan.

[32] Inverse Design of Metamaterials With Adaptable Force–Displacement Characteristics
H. Gökçen Güner, Kaushik Dayal, **Alexandra Ion**.
Journal of Applied Mechanics. July 2026, 93(7): 071006 (8 pages). Paper No: JAM-26-1123
https://doi.org/10.1115/1.4071782

[31] Towards Accessible Mobility Support: User-Centered Design of a Passive, Multi-Functional, Low-Cost Knee Exoskeleton
Yuyu Lin, Yujia Liu, Emma Kim, **Alexandra Ion**.
In Proceedings of ACM CHI'26, Barcelona, Spain. April 13 - 17, 2026. Acceptance rate 25%

[30] Towards Fluent Interaction with Cyber-Physical Architecture
Jesse T Gonzalez, Neeta M Khanuja, Michael Mingxuan Li, Maggie Guo, Layomi Olaitan, Emily Lau, Jenny Pugh, **Alexandra Ion**, Scott E Hudson.
In Proceedings of ACM CHI'26, Barcelona, Spain. April 13 - 17, 2026. Acceptance rate 25%
BEST PAPER AWARD (top 1%)

[29] InSense3D: Designing Smart 3D-Printed Structures Leveraging Ferromagnetic Filaments for Inductive Deformation Sensing
Rahul Bhaumik, Camilo Ayala-Garcia, Niko Münzenrieder, Michael Haller, **Alexandra Ion**.
In Proceedings of ACM CHI'26, Barcelona, Spain. April 13 - 17, 2026. Acceptance rate 25%

[28] Personalized Bistable Orthoses for Rehabilitation of Finger Joints
Yuyu Lin, Dian Zhu, Anoushka Naidu, Kenneth Yu, Deon Harper, Eni Halilaj, Douglas Weber, Deborah Ellen Kenney, Adam J. Popchak, Mark Baratz, **Alexandra Ion**.
In Proceedings of ACM UIST '25. Busan, Republic of Korea. Sept. 28 - Oct. 01, 2025. Acceptance rate: 22%
Best Demo Honorable Mention, Jury's choice.

[27] Towards Unobtrusive Physical AI: Augmenting Everyday Objects with Intelligence and Robotic Movement for Proactive Assistance
Violet Yinuo Han, Jesse T. Gonzalez, Christina Yang, Zhiruo Wang, Scott E. Hudson, **Alexandra Ion**.
In Proceedings of ACM UIST '25. Busan, Republic of Korea. Sept. 28 - Oct. 01, 2025. Acceptance rate: 22%

[26] Transforming Everyday Objects into Dynamic Interfaces using Smart Flat-Foldable Structures
Violet Yinuo Han, Amber Yinglei Chen, Mason Zadan, Jesse T. Gonzalez, Dinesh K. Patel, Wendy Fangwen Yu, Carmel Majidi, **Alexandra Ion**.
In Proceedings of ACM UIST '25. Busan, Republic of Korea. Sept. 28 - Oct. 01, 2025. Acceptance rate: 22%

[25] Sculptable Mesh Structures for Room-Scale Form-Finding
Jesse T. Gonzalez, Yanzhen Zhang, Dian Zhu, Alice Yu, Sapna Tayal, Nazm Furniturewala, Ziying Qi, Somin Ella Moon, Leyi Han, **Alexandra Ion**, Scott E. Hudson.
In Proceedings of ACM UIST '25. Busan, Republic of Korea. Sept. 28 - Oct. 01, 2025. Acceptance rate: 22%

[24] Inductive Pressure Sensors Using 3D-Printed Structures With Tunable Stiffness.
Rahul Bhaumik, Thomas Preindl, **Alexandra Ion**, Camilo Ayala-Garcia, Nitzan Cohen, Michael Haller, Niko Münzenrieder.
IEEE Sensors Letters, vol. 9, no. 5, pp. 1-4, May 2025, Art no. 6005904.

[23] Wearable Material Properties: Passive Wearable Microstructures as Adaptable Interfaces for the Physical Environment. Yuyu Lin, Hatice Gokcen Guner, Jianzhe Gu, Sonia Prashant, **Alexandra Ion.**
Accepted to ACM CHI'25. Acceptance rate: 25%

[22] A Dynamic Bayesian Network Based Framework for Multimodal Context-Aware Interactions
Violet Yinuo Han, Tianyi Wang, Hyunsung Cho, Kashyap Todi, Ajoy Savio Fernandes, Andre Levi, Zheng Zhang, Tovi Grossman, **Alexandra Ion,** Tanya R. Jonker.
In Proceedings of ACM Intelligent User Interfaces (IUI '25).

[21] ConeAct: A Multistable Actuator for Dynamic Materials
Yuyu Lin, Jesse T. Gonzalez, Zhitong Cui, Yash Rajeev Banka, **Alexandra Ion.**
In Proceedings of ACM CHI'24. Acceptance rate: 26%

[20] Robotic Metamaterials: A Modular System for Hands-On Configuration of Ad-Hoc Dynamic Applications
Zhitong Cui, Shuhong Wang, Violet Yinuo Han, Tucker Rae-Grant, Willa Yunqi Yang, Alan Zhu, Scott E. Hudson, **Alexandra Ion.**
In Proceedings to ACM CHI'24. Acceptance rate: 26%

[19] BlendMR: A Computational Method to Create Ambient Mixed Reality Interfaces
Violet Yinuo Han, Hyunsung Cho, Kiyosu Maeda, **Alexandra Ion,** David Lindlbauer
In Proceedings of ACM PACM (ISS'23).
BEST PAPER AWARD.

[18] Reprogrammable Digital Metamaterials for Interactive Devices.
Yu Jiang, Shobhit Aggarwal, Zhipeng Li, Yuanchun Shi, **Alexandra Ion.**
In Proceedings of ACM UIST'23. San Francisco, CA. Oct 29 – Nov 1, 2023. Acceptance rate: 25%

[17] Parametric Haptics: Versatile Geometry-based Tactile Feedback Devices.
Violet Yinuo Han, Abena Boadi-Agyemang, Yuyu Lin, David Lindlbauer, **Alexandra Ion.**
In Proceedings of ACM UIST'23. San Francisco, CA. Oct 29 – Nov 1, 2023. Acceptance rate: 25%

[16] Constraint-Driven Robotic Surfaces, at Human-Scale.
Jesse T Gonzalez, Juhi Kedia, Sapna Tayal, Sonia Prashant, **Alexandra Ion,** Scott E Hudson.
In Proceedings of ACM UIST'23. San Francisco, CA. Oct 29 – Nov 1, 2023. Acceptance rate: 25%
Best demo award.

[15] MiuraKit: A Modular Hands-On Construction Kit For Pneumatic Shape-Changing And Robotic Interfaces.
Zhitong Cui, Shuhong Wang, Junxian Li, Shijian Luo, **Alexandra Ion.**
In Proceedings of ACM DIS'23. Pittsburgh, PA. July 10 – 14, 2023. Acceptance: 24%

[14] Permeable Thermistor Temperature Sensors Based on Porous Melamine Foam
Hugo de Souza Oliveira, Niloofar Saeedzadeh Khaanghah, Violet Yinuo Han, Alejandro Carrasco-Pena, Alexandra Ion, Michael Haller, Giuseppe Cantarella, Niko Münzenrieder.
In IEEE Sensors Letters, Vol. 7, No. 5, pp. 1-4, Art no. 2500904, May 2023

[13] Reconfigurable Elastic Metamaterials.
Willa Yunqi Yang, Yumeng Zhuang, Luke Darcy, Grace Liu, **Alexandra Ion**.
In Proceedings of ACM UIST '22. Bend, Oregon. Oct 29 – Nov 2, 2022

[12] Developable Metamaterials: Mass-fabricable Metamaterials by Laser-Cutting Elastic Structures.
Madlaina Signer, **Alexandra Ion**, and Olga Sorkine-Hornung.
In Proceedings of ACM CHI ’21. Virtual Conference (originally Yokohama, Japan), May 8 – 13, 2021.

[11] Shape Approximation by Developable Wrapping.
Alexandra Ion, Michael Rabinovich, Philipp Herholz, and Olga Sorkine-Hornung.
ACM Transactions on Graphics 39, 6. 2020. (Proceedings of SIGGRAPH ASIA)

[10] Understanding Metamaterial Mechanisms.
Alexandra Ion, David Lindlbauer, Philipp Herholz, Marc Alexa, and Patrick Baudisch.
In Proceedings of ACM CHI ‘19. Glasgow, UK, May 4 – 9, 2019.

[9] TrussFormer: 3D Printing Large Kinetic Structures.
Robert Kovacs, **Alexandra Ion**, Pedro Lopes, Tim Oesterreich, Johannes Filter, Philipp Otto, Tobias Arndt, Nico Ring, Melvin Witte, Anton Synytsia, and Patrick Baudisch.
In Proceedings of ACM UIST’18. Berlin, Germany, October 14 – 17, 2018.
ACM CHI’19 Video Showcase **Best Visual Communication Award.**

[8] Metamaterial Textures.
Alexandra Ion, Robert Kovacs, Oliver Schneider, Pedro Lopes and Patrick Baudisch.
In Proceedings of ACM CHI ‘18. Montreal, Canada, April 21 – 26, 2018.

[7] Adding Force Feedback to Mixed Reality Experiences and Games using Electrical Muscle Stimulation.
Pedro Lopes, Sijing You, **Alexandra Ion**, and Patrick Baudisch.
In Proceedings of ACM CHI ‘18. Montreal, Canada, April 21 – 26, 2018.

[6] Digital Mechanical Metamaterials.
Alexandra Ion, Ludwig Wall, Robert Kovacs, and Patrick Baudisch.
In Proceedings of ACM CHI ‘17. Denver, CO, May 6 – 11, 2017.

[5] Metamaterial Mechanisms.
Alexandra Ion, Johannes Frohnhofen, Ludwig Wall, Robert Kovacs, Mirela Alistar, Jack Lindsay, Pedro Lopes, Hsiang-Ting Chen, and Patrick Baudisch.
In Proceedings of ACM UIST’16. Tokyo, Japan, October 16 – 19, 2016. pp. 529-539.
BEST PAPER HONORABLE MENTION (top 2% of submissions).

[4] Skin drag displays: dragging a physical tactor across the user’s skin produces a stronger tactile stimulus than vibrotactile.
Alexandra Ion, Edward Wang, and Patrick Baudisch.
In Proceedings of ACM CHI’15. Seoul, Korea, April 18 – 23, 2015. Short paper.

[3] Proprioceptive Interaction.
Pedro Lopes, **Alexandra Ion**, Willi Mueller, Daniel Hoffmann, Patrik Jonell, Patrick Baudisch.
In Proceedings of ACM CHI’15. Seoul, Korea, April 18 – 23, 2015.

[2] Impacto: Simulating Physical Impact by Combining Tactile Stimulation with Electrical Muscle Stimulation.
Pedro Lopes, **Alexandra Ion**, Patrick Baudisch.
In Proceedings of ACM UIST’15. Charlotte, NC, November 8 – 11, 2015.

[1] Canyon: Providing location awareness of multiple moving objects in a detail view on large displays.
Alexandra Ion, Yu-Ling Chang, Michael Haller, Mark Hancock, Stacey Scott.
In Proceedings of ACM CHI’13. Paris, France, April 27 – May 2, 2013. 3149-3158.
BEST PAPER HONORABLE MENTION (top 5% of submissions).

NON-PEER-REVIEWED PUBLICATIONS

[6] Bridging disciplines for a new era in Physical AI
Alexandra Ion, Carmel Majidi, Lining Yao, Amir Alavi. ACM UIST 2024 Workshop.

[5] Interactive Metamaterials.
Alexandra Ion and Patrick Baudisch.
Magazine article. ACM interactions, January/February 2020.

[4] Metamaterial Devices.
Alexandra Ion and Patrick Baudisch.
In ACM SIGGRAPH 2018 Studio.

[3] Using Your Own Muscles: Realistic physical experiences in VR.
Pedro Lopes, **Alexandra Ion**, Robert Kovacs.
Magazine article, ACM XRDS, Fall 2015, Vol. 22.

[2] Understanding mid-air hand gestures: A study of human preferences in usage of gesture types for HCI.
Roland Aigner, Daniel Wigdor, Hrvoje Benko, Michael Haller, David Lindlbauer, **Alexandra Ion**, Shengdong Zhao, and Jeffrey Tzu Kwan Valino Koh.
Technical Report MSR-TR-2012-111. 2012.

[1] Hot Topics in Personal Fabrication Research. Tutorial.
Stefanie Mueller, **Alexandra Ion**, and Patrick Baudisch.
In *Proceedings of ACM ITS 2014, 499-502.*

DEMONSTRATIONS

…of physical prototypes and software during hands-on sessions

[12] Bistable finger orthosis. *CMU XRTC Symposium 2025 Demonstration.*
[11] Unobtrusive Physical AI. *CMU XRTC Symposium 2025 Demonstration.*
[10] Bistable finger orthosis. *ACM UIST 2025 Demonstration.* ***Best Demo Honorable Mention, Jury's choice.***
[9] Unobtrusive Physical AI. *ACM UIST 2025 Demonstration.*
[8] Constraint-Driven Robotic Surfaces. *ACM UIST 2023 Demonstration.* ***Best demo award.***
[7] Parametric Haptics. *ACM UIST 2023 Demonstration.*
[6] Reprogrammable Digital Metamaterials for Interactive Devices. *ACM UIST 2023 Demonstration.*
[5] Metamaterial Devices. *ACM SIGGRAPH 2018 Studio Installation. August 2018.*
[4] Metamaterial Devices. *ACM CHI 2018 Interactivity. May 2018.*
[3] Metamaterial Mechanisms & Digital Mechanical Metamaterials. *ACM CHI 2017 Interactivity. May 2017.*
[2] Metamaterial Mechanisms. *ACM UIST 2016 Demonstration. October 2016.*
[1] Impacto: Simulating Physical Impact by Combining Tactile Stimulation with EMS. *ACM UIST 2015 Demonstration. November 2015.*

Awards & Honors

ACADEMIC

[13] **Best paper** award. ACM CHI 2026.
[12] **NSF CAREER** award 2025.
[11] Best Demo Honorable Mention, Jury's choice. ACM UIST 2025.

[10] **Best paper** award. ACM PACM (ISS'23).
[9] Best demo award. ACM UIST 2023.
[8] Received the **Sara Kiesler Junior Development Fellowship** at CMU HCII for 2023-2024**.**
[7] Selected for **Rising Stars in EECS 2019.**
[6] Best visual communication award for TrussFormer. ACM CHI 2019 Video Showcase.
[5] Scholarship for academic exchange (2018 DAAD FIT program, ~9 000 EUR)
[4] My CHI 2015 talk is the example for a good presentation on the CHI conference website, 2016 – present
[3] **Best paper honorable mention** award. ACM UIST 2016.
[2] **Best paper honorable mention** award. ACM CHI 2013.
[1] Nominated for national best master's thesis award. 2012.

INDUSTRY/ART/DESIGN

[5] Fast Company's Innovation by Design Award (Honorable Mention) 2018, for Adding Force Feedback to MR.
[4] VIDA. Incentive Award 2016, for Ad Infinitum.
[3] Bronze Cyber Lion, Cannes Lions 2010, for "Last call" (interactive movie)
[2] Bronze Media Lion, Cannes Lions 2010, for "Last call" (interactive movie)
[1] Bronze "Digital Media Craft", ADC Germany 2010, Art directors club, for "Last call" (interactive movie)

Invited Talks & Panels

[51] CMU Turn Tartan: Women in STEM Panel Discussion. April 23, 2026.
[50] CMU Center for Collaboration Science, panelist "Expert Panel on Collaboration in Action". February 13, 2026.
[49] CMU Lab to Market Spring 2026, panelist "Research Outlook". February 11, 2026.
[48] Princeton University, hosted by Parastoo Abtahi. October 27, 2025.
[47] Playing Models conference, invited speaker. August 13, 2025.
[46] Materials Research Society 2024 Fall Meeting. December 2, 2024.
[45] Extended Reality Technology Center. November 21, 2024.
[44] University of Konstanz, hosted by Tiare Feuchtner & Harald Reiterer. July 6, 2023.
[43] Austrian CS Day, hosted by IST Austria. June 21, 2022.
[42] University of Washington, Computational Fabrication, hosted by Adriana Schulz, Jeff Lipton. April 13, 2022.
[41] Stony Brook University, Civil Engineering Seminar, hosted by Paolo Celli. November 15, 2021.
[40] Tsinghua University, Access Computing Summer Program, hosted by Yukang Yan. August 10, 2021.
[39] MIT Summer Geometry Institute, hosted by Justin Solomon. August 3, 2021.
[38] Stanford HCI Seminar, hosted by Michael Bernstein. May 28, 2021.
[37] Toronto Geometry Colloquium, hosted by Alec Jacobson, Silvia Sellan. March 31, 2021.
[36] CMU Mechanics Seminar, hosted by Kaushik Dayal. October 2, 2020.
[35] University of Toronto, hosted by Tovi Grossman. March 26, 2020.
[34] Carnegie Mellon University, hosted by Scott Hudson. March 23, 2020.
[33] Boston University MechE, hosted by Alice White. March 17, 2020.
[32] Boston University CS, hosted by Emily Whiting. March 16, 2020.
[31] University of Chicago, hosted by Blase Ur. March 12, 2020.
[30] Aalto University, hosted by Elisa Mekler, Antti Oulasvirta. March 5, 2020.
[29] IST Austria, hosted by Bernd Bickel. March 2, 2020.
[28] Cornell University, hosted by Cheng Zhang, François Guimbretière. February 26, 2020.

[27] TU Delft CS, hosted by Elmar Eisemann. Februray 7, 2020.
[26] TU Delft PME, hosted by Just Herder. Februray 4, 2020.
[25] Aalto University, hosted by Antti Oulasvirta. December 17, 2019.
[24] University of Copenhagen. December 16, 2019
[23] TU Delft, hosted by Just Herder. November 14, 2019.
[22] University of Chicago, hosted by Pedro Lopes. October 29, 2019.
[21] University of Stuttgart, hosted by Achim Menges. September 16, 2019.
[20] Google (Mountain View), hosted by Alex Olwal. August 9, 2019.
[19] UC Berkeley, hosted by Björn Hartmann. August 8, 2019.
[18] Stanford, hosted by Sean Follmer. August 7, 2019.
[17] UCLA, hosted by Ankur Mehta. August 2, 2019.
[16] MIT Media Lab, hosted by Hiroshi Ishii. June 19, 2019.
[15] MIT CSAIL, hosted by Stefanie Mueller. June 19, 2019.
[14] ETH Zurich, PhD Seminar. October 26, 2018.
[13] University of Sussex, hosted by Diego Martínez & Sriram Subramanian. May 21, 2018.
[12] Université de Montréal, hosted by Bernhard Thomaszewski. April 27, 2018.
[11] IST Austria, hosted by Bernd Bickel. March 2, 2018.
[10] ETH Zürich, hosted by Olga Sorkine-Hornung. December 14, 2017.
[9] SAP Walldorf, hosted by Andreas Polze & Bernd Welz. December 11, 2017.
[8] TU Delft, hosted by Jouke Verlinden. December 4, 2017.
[7] Aarhus University, hosted by Roman Rädle. October 5, 2017.
[6] Technomania, hosted by The Danish Society of Engineers IDA. October 4, 2017
[5] Driving 3D, hosted by The Danish Society of Engineers, IDA. September 29, 2017.
[4] TEDx Poznan, Poland. April 8, 2017.
[3] Singularity University Copenhagen, hosted by Märtha Rehnberg. December 14, 2016.
[2] TU Berlin, hosted by Marc Alexa. 2016.
[1] TU Berlin, hosted by Marc Alexa. 2015.

Professional Service

PROGRAM COMMITTEE

Program committee **chairing**:	ACM **UIST 2024** program co-chair ACM **CHI 2024** subcommittee co-chair “Building Devices” ACM SIGGRAPH 2024 – Poster chair ACM **CHI 2023** subcommittee co-chair “Building Devices”
Program committee member:	ACM **CHI** 2022, 2021, 2020, 2019 ACM **UIST** 2026, 2022, 2021, 2020 ACM CHI Best paper committee 2023, 2022 ACM SIGGRAPH 2023, 2022, 2021 – Posters committee ACM SIGGRAPH Asia 2019 – Technical briefs and posters Graphics Interface 2020

CONFERENCE ORGANIZING

Organizing committee:	ACM UIST 2027 – Local Experience chair ACM SIGGRAPH 2023 – Student Research Competition chair ACM UIST 2022, 2023 – Doctoral Symposium co-chair ACM UIST 2020, 2021 – Demonstrations co-chair ACM UIST 2019 – Sustainability co-chair ACM UIST 2018 – Local arrangements co-chair, Accessibility chair ACM UIST 2017 – Student volunteer co-chair ACM UIST 2016 – Documentation chair
Session chair:	at ACM CHI 2026, session "Interactive Design Systems for Fabrication" at ACM CHI 2023, session "Wearables and Materials" at ACM CHI 2019, session "Designing with Materials" at ACM CHI 2018, session "Craft, Fabrication, Making" at ACM CHI 2017, session "Haptics on Skin"
Student volunteer:	ACM CHI 2016, ACM UIST 2015, ACM ITS 2014

REVIEWING

Conferences:	ACM CHI 2018 – 2014 ACM UIST 2025, 2023, 2019 – 2016 ACM SIGGRAPH 2024, 2023, 2020 ACM SIGGRAPH Asia 2018 ACM PACM (ISS) 2026 ACM TEI 2024, 2019, 2017 ACM DIS 2023, 2017, 2016 ACM Augmented Human 2016, 2015 IEEE World Haptics 2015, 2017 IEEE Haptics Symposium 2018 ACM ICMI 2017, ACM SUI 2016
Journals:	Programmable Materials 2023 ACM Transactions on Graphics 2022 ACM IMWUT 2019 Elsevier Computers & Graphics 2018 Nature Scientific Reports 2018 Interacting with Computers 2014

UNIVERSITY/DEPARTMENT SERVICE

2026	Interim Associate Director of HCI Undergraduate Programs
2026 - present	Advisor to the PhD program at the Free University of Bozen-Bolzano, Italy
2026	HCII MHCI admissions committee
2024/2025	HCII PhD admissions co-chair
2023 – now	HCII Executive committee

2022/2023	HCII PhD admissions co-chair
2021/2022	HCII PhD admissions committee
2022	HCII REU admissions committee
2021	Giving HCII MHCI capstone critique
2021	HCII BHCI admissions committee
2021	HCII REU admissions committee

OUTREACH

2022 – 2025	Mentor for WiGRAPH (ACM Community for Women in Computer Graphics Research) providing mentorship for early career women in computer graphics & HCI
2022 – 2023	Mentor for junior CMU HCII PhD students leading a mentoring group to support junior PhD students, bi-weekly meetings
2022 – 2023	Lecturer at the Computational Interaction summer schools (Saarbrücken and Ann Arbor). Topic: “Geometric optimization” (3-hour lecture)
2023	Hosting 1 student for their Research Experience for Undergraduates (REU) at HCII, CMU
2022 – 2023	External mentor for Research Experience for Undergraduates (REU) Site at the University of Texas at Arlington, topic: “Hybrid Design and Fabrication”
2021	Guest speaker at MIT Summer Geometry Initiative a research-oriented summer school for international undergraduate students
2021	Guest speaker at Tsinghua Global Innovation Exchange Institute a research-oriented summer school for international undergraduate students
2021	Hosting 2 students for their Research Experience for Undergraduates (REU) at HCII, CMU
2019 – 2020	Co-lead of the Network of Women in Computer Science (CSNOW) at ETH Zurich an organization to mentor women in CS, provide mentoring and networking opportunities.

Advising & mentoring

PHD STUDENTS

Sep 2025 - present	Yujia Liu PhD student at CMU HCII
Sep 2023 – present	Violet Han PhD student at CMU HCII
Sep 2022 – present	Yuyu Lin PhD student at CMU HCII

Sep 2021 – 2026	Hatice Gökçen Güner PhD student at CMU CEE Co-mentored with Kaushik Dayal (CMU CEE)
Sep 2019 – 2025	Jesse Gonzalez PhD student at CMU HCII Co-advised with Scott Hudson (CMU HCII)

MASTER'S THESES

2025 - 2026	Ziru Wei (CMU SoA) Context-Aware Approach for Proactive Robots. (submitted to ACM UIST 2026)
2022 – 2023	Violet Han (CMU SoA) Parametric Haptics. (accepted at ACM UIST 2023)
2021 – 2022	Yunqi Willa Yang (CMU SoA) Robotic Metamaterials. (accepted at ACM UIST 2022)
2020	Madlaina Signer (ETH Zurich) Developable Metamaterials: Mass-fabricable Metamaterials by Laser-Cutting Elastic Structures. (accepted at ACM CHI 2021)
2016	Ludwig Wilhelm Wall (Hasso Plattner Institute, Germany) Design and Synthesis of Digital Mechanical Metamaterials. (accepted at ACM CHI 2017)

THESIS COMMITTEES

Jesse T. Gonzalez. Carnegie Mellon University. November 14, 2025.
PhD Thesis “Room-Scale Robotic Surfaces”

Jianzhe Gu. Carnegie Mellon University. November 1, 2024.
PhD Thesis “Computational Design of Morphing Looped Graph Structures”

Velko Vechev. ETH Zurich. February 5, 2024.
PhD Thesis “Computational design and fabrication of wearable haptics”

Semyon Efremov. Université de Lorraine. May 3, 2021.
PhD Thesis “Parametrized growth and procedural noise for mechanical metamaterial design”

Teaching

FULL COURSES

Spring 2026	Instructor for 05-391/891 “Designing Human-Centered Software” CMU. 65 students
Fall 2025	Instructor for 05-391/891 “Designing Human-Centered Software” CMU. 76 students
Spring 2025	Instructor for 05-499B/899B “Computational Methods for Interactive Systems. CMU.
Fall 2024	Instructor for 05-391/891 “Designing Human-Centered Software” CMU. 44 students
Fall 2023	Instructor for 05-430/630 “Programming Usable Interfaces”. CMU. ~87 students.

Spring 2023	Instructor for 05-435/835 “Applied Fabrication for HCI”. CMU. ∼33 students.
Fall 2022	Instructor for 05-430/830 “Programming Usable Interfaces”. CMU. ∼92 students.
Spring 2022	Instructor for 05-435/835 “Applied Fabrication for HCI”. CMU. ∼22 students.
Fall 2021	Instructor for 05-430/830 “Programming Usable Interfaces”. CMU. ∼102 students.

GUEST LECTURES

Fall 2025	07-300 Research and Innovation in Computer Science (CMU CS)
Spring 2021, 2022, 2024	05-120 Introduction to Human-Computer Interaction (CMU HCII)
Fall 2023	05-499 Interactive Extended Reality, 2 lectures on “Geometric Optimization” (CMU HCII)
Spring 2021	05-300 HCI Undergraduate Pro Seminar (CMU HCII)

Research Grants & Gifts

2025 – 2030 **NSF CAREER: Material-Changing Interfaces**
Funding body: National Science Foundation
Investigators: PI: Alexandra Ion
Amount: $650,000

2025 – 2026 **Detecting Sensory Mismatch and Alignment through EEG**
Funding body: BMW research
Investigators: PIs David Lindlbauer, Alexandra Ion, Sossena Wood
Amount: $150,000

2023 – 2024 **Low-cost trial prosthetic feet for at home testing**
Funding body: Pennsylvania Infrastructure Technology Alliance
Investigators: PI: Alexandra Ion, Co-PI: Douglas Weber, Industry partner: Humotech
Amount: $81,358

2022 – 2023 **Self-reconfigurable haptic actuation for multi-modal metaverse**
Funding body: Accenture Technology Labs
Investigators: PIs: Alexandra Ion, David Lindlbauer, Nik Martelaro
Amount: $110,000

2021 – 2022 **Custom & Lightweight Prosthetics Using Metamaterials**
Funding body: CMU CIT IS4 Moonshot
Investigators: Alexandra Ion, Kaushik Dayal
Amount: $85,122

FELLOWSHIPS & AWARDS FOR MY ADVISEES

2026 – 2027 “Building Helpful Embodied AI Agents with Long Horizon Understanding”
Qualcomm Innovation Fellowship (North America) 2026
for **Violet Han** (team with Wei Wei Sun)

2024 – 2025 “3D Printing Paralysis Orthotics with Microstructures for Sensing and Actuation”
CMLH Fellowship in Digital Health Innovation
for **Yuyu Lin**

2023 – 2024 “Machine Learning Driven Metamaterial Design for Customizable Prosthetic Sockets with Real-Time High-Fidelity Sensors”
CMLH Fellowship in Digital Health Innovation
for **Hatice Gokcen Guner**

Invited Exhibitions

METAMATERIAL DEVICES

Ars Electronica Center, Linz, Austria. 2019 – present.
BMBF (**German federal ministry for education and research**) InnoTruck (travelling exhibition). 2017 – 2020.
UNIVERSUM Mexico City, “3D, printing the world”. September 2019 – February 2020.
Espacio Fundación Telefónica Chile, Santiago, “3D, printing the world”. March 2019 – July 2019.
Espacio Fundación Telefónica Argentina, Buenos Aires, “3D, printing the world”. July 2018 – October 2018.
Espacio Fundación Telefónica Peru, Lima, “3D, printing the world”. December 2017 – April 2018.
Espacio Fundación Telefónica Spain, Madrid, “3D, printing the world”. June – September 2017.
CeBit 2017. Futurium booth (Berlin-based museum for ideas of the future). March 20 – 24, 2017.

AD INFINITUM (http://a-parasite.org/)

Przemiany Festival, Copernicus Science Center Warsaw. September 13 – 16, 2018.
World Economic Forum, San Francisco. December 2017 – May 2018.
Ars Electronica Festival, Linz, Austria. 7-11 September 2017.
Science Gallery Dublin, Ireland, “Humans need not apply”. February – May 2017.
Natural History Museum Bern, Switzerland. September 9, 2016.

Selected Press

CNET. What If Your Coffee Mug Knew Your Next Move? AI Researchers Made It Happen. 2026.
CMU News. CMU Researchers Use AI To Turn Everyday Objects Into Proactive Assistants. 2025.
3dprint.com. 3D Printed Customizable Bistable Finger Orthotic for Arthritis & Injury Recovery. 2025.
CMU News. CMU Researchers Develop Customizable Finger Brace for Injury Recovery. 2025.

IEEE Spectrum. Mechanical Metamaterials and other 3D Printing Tech from CHI 2017. 2017.
3ders.org. Researchers use 3D printing to make 'digital mechanical metamaterials' that function like machines. 2017.
think3D. ‘Digital Mechanical Metamaterials’ made by the Researchers. 2017.
digital trends. This 3D printed door latch can be unlocked with a PIN code, yet doesn’t require electricity. 2017.

fast company. These Metamaterials Act Like Machines. 2016.
gizmodo. This Simple 3D-Printed Door Handle Works Without Any Moving Parts. 2016.
creative applications. Metamaterial Mechanisms – 3D Grids with Mechanical Properties. 2016.
Make:. “Metamaterials” Allow You to 3D Print Simple Machines. 2016.
dezeen. Metamaterials Make it Possible to Create Mechanisms from a Single Piece of Plastic. 2016.
digital trends. 3D-printed Metamaterial ‘Machines’ are Greater than the Sum of their Parts. 2016.

Previous professional experience

Mar 2013 - Jul 2013	Software Developer & Usability Engineer (Full time employee, Professional/Experienced) Bernecker+Rainer Industrie Elektronik Ges.m.b.H., http://www.br-automation.com/
Sep 2010 - Feb 2012	Software Developer (Part time employee, Professional/Experienced) Interactive Pioneers GmbH, http://interactive-pioneers.com/ (former Powerflasher GmbH)
Oct 2009 - Aug 2010	Software Developer (Full time employee, Entry Level) Interactive Pioneers GmbH, http://interactive-pioneers.com/ (former Powerflasher GmbH) Developer for WPF, Silverlight, Java Backend, Actionscript 3.0, Flex
Mar 2009 - Aug 2009	WPF/Silverlight Developer (Student/Intern) Interactive Pioneers GmbH, http://interactive-pioneers.com/ (former Powerflasher GmbH)
Aug 2008 - Sep 2008	Screen/Web Designer, Web Developer (Student/Intern) Lomographic Society International, http://www.lomography.com/